

2026-08-11 Issues when constructing the analysis sample

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Outline

① Two analysis samples

② A weighted sample of job postings

The analysis sample I: New hires

The sample of new hires (definitive and intuitive):

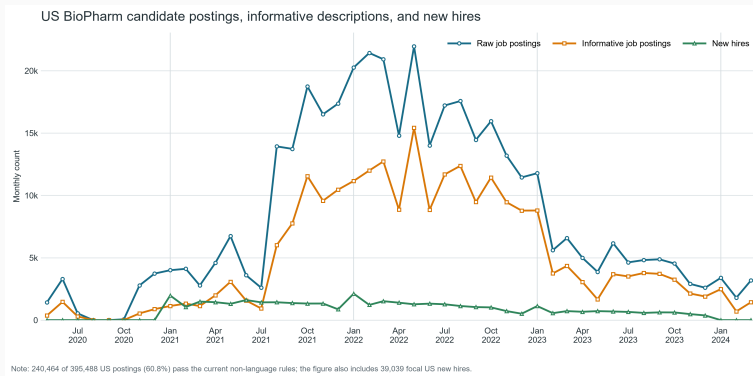
- The construction process:
 - ① Retain employment spells in the focal industry-occupation cells.
 - ② Further retain employment spells starting from Jan 2021 to Dec 2023.
 - ③ Exclude employment spells with missing geography or job title information.
 - ④ Exclude internship positions.
 - ⑤ Retain one employment within each user-company cell (the [final sample of focal new hires](#)).
- Interpretations:
 - It contains one row per user-company cell; the same user can appear at multiple companies.
 - It is not necessarily the user's first spell at that company: internal transfers into a focal occupation are intentionally retained (e.g., if an accountant becomes a scientist within the same company, then I keep only his/her scientist employment spell and count as a new hire).

The analysis sample II: Job postings

The sample of (LinkedIn) job postings (problematic and uncertain):

- Background information about the raw job postings data:
 - Each row is one unique job posting.
 - Most rows have non-missing company id, job title variable, and Revelio standardized ONET occupation classification.
 - The variable `description` stores the full job advertisement texts scraped by Revelio.
 - Many of the description texts are not informative. There could be multiple reasons behind this.
 - The original job posting is not informative (e.g., containing only a url link or email address).
 - Revelio may fail to scrape the full texts successfully (e.g., some texts are clearly a partial fragment of the full job advertisements).
- Why cannot I replicate what I did before to construct the sample of job postings?
 - I will name the sample of job postings as the **sample of candidate job postings** that is constructed in the same way as the final sample of focal new hires. ▶ Construction
 - There are substantially more candidate job postings than the final sample of focal new hires.
 - It doesn't make sense to weight each job posting equally.

There are substantially more candidate job postings than the final sample of focal new hires.



- In the BioPharm industry, from 2021-23, in US, there are 39,039 focal new hires (the green line).
- Within a corresponding time window for job postings, in the BioPharm industry, in US, there are 395,488 candidate job postings (the blue line). There are 10 times more job postings than the number of focal new hires.
- Even if I focus on only those **informative** job postings, there are substantially more informative job postings than new hires (240,464 vs. 39,039). [▶ Construction](#)

Outline

- ① Two analysis samples
- ② A weighted sample of job postings

A more reasonable way to construct the sample of job postings

- My original thought is that we match each new hire with one (and only one) job posting that is mostly likely to be the corresponding job posting.
 - However, this will introduce some endogenous sample selection issues – it is very hard to match a new software engineer at Google to the corresponding job posting.
 - Gaurav says that we don't need to do this matching at all. This is not so reasonable because with 10 times more job postings, we need to somehow restrict the space of job postings to match the number of new hires.
- A more reasonable way is to weight job postings with the final sample of new hires.
 - We restrict the attention to the sample of informative candidate job postings because we can construct a series of indicators measuring the level of selectivity (education/experience/skill requirements, etc). [▶ More justifications](#)
 - Starting from the final sample of new hires, we match each new hire with multiple job postings based on ONET occupation classification and posting date.
 - The matched job postings can have different weights (e.g., we can assign a higher weight to a job posting that is closer to the new hire's start date).
 - But the sum of weights across all job posting matched to one new hire will always be 1.
 - For each new hire, using all mapped candidate job postings, we can construct a weighted average level of selectivity facing by each new hire.
 - In this way, we can reach a balanced sample of new hires + job postings.
 - The interpretation of the weighted average of selectivity indicators is that we measure the company-level average ex ante selectivity a new hires faces before he joins the company.

Next steps

- I'd like to do more work on the deterministic rule-based algorithm (more validation and tests) to identify those informative job postings because they are clearly a better subset of job postings of interest.
- I'd like to do more work in using LLM to construct a series of indicators measuring the level of selectivity with the job postings.
 - Surprisingly, I don't think there is any economics paper trying to build a reasonable pipeline for this task.
 - This could be one of our contributions too – the way to measure selectivity from job advertisements.
- I would say, the construction of the analysis sample (especially for the job postings sample) is more like an art. In my opinion, a weighted sample of job postings based on the sample of new hires is better to keep all job postings (even the uninformative ones) and create a simple average of selectivity indicators across all job postings.

More on the analysis sample: At which level of aggregation

- After determining the sample of new hires and job postings, we need to determine the level of aggregation. That is to say, is our revealed hiring practice measure at company level, or company-year level, or company-state level, or company-state-year level.
- This is partly a data-driven process, because within each cell, we need a minimum number of new hires to make our measurement meaningful.
- This will also be based on implications/validation tests we want to have. In some earlier discussion, we talk about using mergers/IPOs as shocks to examine if companies change their hiring practices after these events, then the level of aggregation has to be at least company-year level.

Outline

③ Appendix

The sample of candidate job postings

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- 1 Construct one extraction window per focal company. For each exact company identifier, extend from 18 months before its first focal hire through 3 months after its last focal hire.
- 2 Retain job postings within those companies that have at least 1 focal new hire.
- 3 Retain job postings within the same focal ONET occupations.
- 4 Retain job postings within the Step-1 constructed company-specific posting-date windows.
- 5 Exclude internship advertisements by specified job titles.

The sample of candidate informative job postings

▶ Back

- ➊ Hash exact descriptions so repeated text is processed once, while retaining every posting in the final sample.
- ➋ Parse HTML or plain text into ordered blocks, normalize the text conservatively, and hash each distinct normalized description.
- ➌ Keep a description only if it has at least 600 visible characters, 100 words, three informative units, two substantive content areas, and duties or qualification content. Exclude boilerplate-dominated, corrupted, noncurrent, multi-role, or truncated text.
- ➍ Map the description-level decision back to every posting. This identifies 240,464 informative postings, or 60.8% of the 395,488 US candidate postings.

Focusing on a cleaner subset of informative job postings

► Back

- We have substantially more job postings than new hires, and we need to reduce the sample size of job postings (at least to a magnitude that is similar to the number of new hires).
- Uninformative job postings could mean
 - the original job posting is not informative, or
 - Revelio fails to scrape the full job advertisements (some description texts are clearly only a partial fragment of full texts); and we don't know why they are uninformative.
- It is always good to start with a cleaner subsample for analysis, and use the full sample for robustness.